

Electricity

We use electricity to power all sorts of things, from factories and trains to the many small appliances in our homes. The energy it contains comes from charged electrons that whizz around inside every atom.

ELECTRICITY IN NATURE

Electricity is not only generated in power stations – it is also found in nature, from high-energy lightning strikes to inside our own bodies. Our brains use electric signals to tell our muscles to move.



LIGHTNING

A bolt releases as much energy as a power station makes in one second.



AURORA

These lights in the sky are streams of electrically charged particles.



NERVOUS SYSTEMS

Human nerves communicate by electric signals.

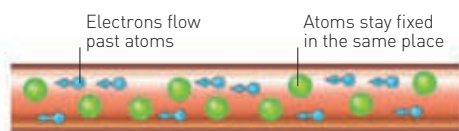


ELECTRIC EEL

This eel discharges electricity in water to kill fish for food.

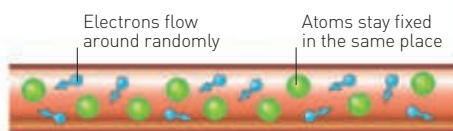
ELECTRIC CURRENT

When electrons flow down wires, they carry energy from place to place. So in a torch, electrons march around the wire from the battery to the lamp, where their power lights up the bulb.



CURRENT FLOWING

When the power is switched on, the electrons move along in a line, forming an electric current.



NO CURRENT FLOWING

When the power is switched off, there's nothing to move the electrons in a line, so they just jig about randomly.

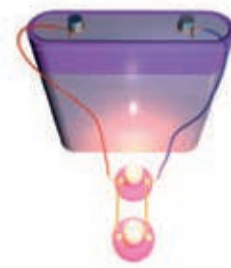
CIRCUITS

The path that electrons travel along is called a "circuit". A circuit carries power from a power source (such as a wall socket) to something that needs electricity to run (such as a lamp). There are two types of circuit.



SERIES CONNECTION

All the power moves through each part of the circuit, in a line.

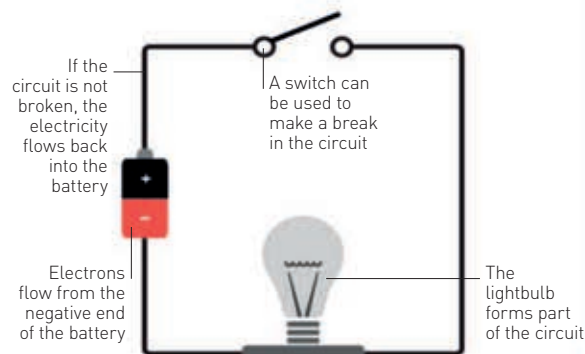


PARALLEL CONNECTION

The power splits into two as it reaches two lamps wired like this.

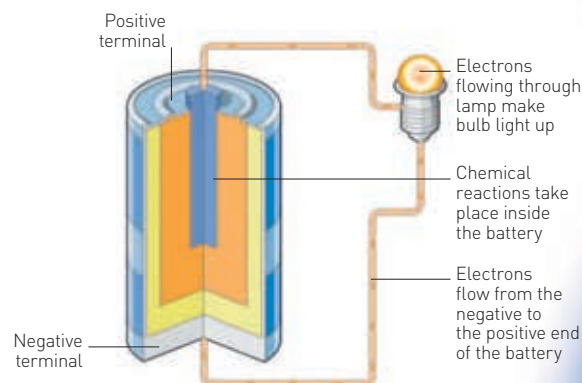
SWITCHES

If you attach a wire to both ends of a battery, and connect a lightbulb to the wire at some point, the electricity would continually flow and always light the bulb. A switch is used to break the circuit, so the bulb can be switched on and off.



BATTERIES

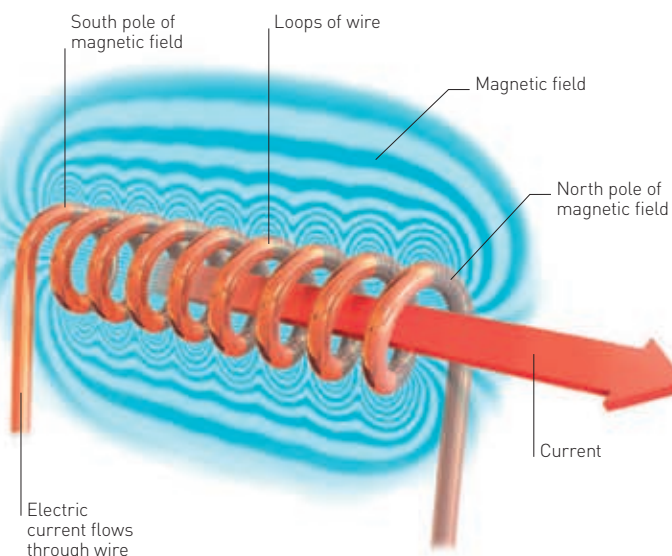
Batteries make their own electricity by using chemicals. When you connect a battery, chemical reactions take place that generate electrons.



LIGHTNING BOLTS ARE LARGE-SCALE STATIC SHOCKS

ELECTROMAGNETISM

When an electric current flows through a wire, it creates a magnetic field around it. The strength of the magnetic field can be increased by coiling the wire in loops, because that allows more current to flow through a smaller distance.



STATIC ELECTRICITY

Static electricity is sometimes created when two things are rubbed together. The rubbing creates an electrical charge, which is released when it comes into contact with something else that conducts electricity.

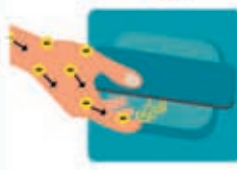
GETTING A SHOCK

Static shocks occur because your body builds up static when you rub against things. The static stays until you touch something metal, when it moves from you through the metal to Earth, giving you a shock.



1 CHARGED UP

The electrical charge you pick up from rubbing against things is negative. It will stay in your body as you move around, until you touch a positively charged object such as a metal handle.



2 JUMPING ELECTRONS

When you touch a conductor, such as a metal handle, the static charge jumps from you, to the handle, to Earth. As the negatively charged electrons jump across, you feel a static shock.